FIITJEE

ALL INDIA TEST SERIES

FULL TEST – X

JEE (Main)-2023

TEST DATE: 31-03-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. A

Sol. By Kepler's 2nd law

$$\frac{\Delta A}{\Delta t} = \frac{L}{2m} = \frac{\pi ab}{T}$$

$$L = \frac{m\pi(r_a + r_p)\sqrt{r_a r_p}}{T}$$

2. D

Sol. $$\tau = \frac{mgL}{2} = I\alpha$$

$$\frac{ML^2}{3}\alpha = \frac{MgL}{2}$$

$$\alpha = \frac{3g}{2L}$$

3. D

Sol. $T = 2ma_C$ …(i)

$$TR = \frac{1}{2}(2m)R^2\alpha \quad \text{…(ii)}$$

$$J = mv \Rightarrow v = \frac{J}{m} = \sqrt{10}$$

$$T + \frac{T}{2} + T = \frac{mv^2}{2R} \quad \text{…(iii)}$$

For More Material Join: @JEEAdvanced_2024

$$a_C = \frac{T}{2m} = \frac{v^2}{10R} = 4\ m/s^2$$

4. C

Sol. $V_1 = V_4 = 0$

$$\frac{Q-q}{A\varepsilon_0}d - \frac{q}{A\varepsilon_0}2d - \frac{q}{A\varepsilon_0}d = 0$$

$$q = \frac{Q}{4}$$

5. C

Sol. Let stone hit the wall at height h

$$h = H - \frac{1}{2}gt^2 \quad \ldots(i)$$

$$H = \frac{1}{2}g(2t)^2 \quad \ldots(ii)$$

Solving equation (i) and equation (ii), $h = \frac{3H}{4}$

6. B

Sol. $v_{||}$ to wall remains constant

$v_{\perp}$ to wall becomes $\left(-\frac{1}{2}\right)2\hat{i} = -\hat{i}$

$$\Rightarrow \vec{v}_f = -\hat{i} + 2\hat{j}$$

7. A

Sol. $T = \frac{Mg}{2}$

$$mg - \frac{Mg}{2} = ma \quad \ldots(i)$$

$$\frac{Mg}{2} - m'g = m'a \quad \ldots(ii)$$

From equation (i) and (ii)

$$\frac{4}{M} = \frac{1}{m} + \frac{1}{m'}$$

8. C

Sol. $a_{system} = \frac{g}{4}$

For block A not to slip on B, we have $\frac{mg}{4} = \mu mg$

$$\Rightarrow \mu = \frac{1}{4}$$

9. B

Sol. For equilibrium

$$I\left(\frac{a}{2}\right)B = Mg$$

For More Material Join: @JEEAdvanced_2024

10. C

Sol. $\tan\phi = \frac{x_L}{R} = 1 \Rightarrow x_L = R$

After addition of capacitor, power factor becomes

$x_L = x_C = \frac{1}{\omega C}$

$Z\,(\text{Box}) = \sqrt{R^2 + x_L^2} = \frac{\sqrt{2}}{\omega C}$

11. C

Sol. $T = T_0 e^{\alpha V}$

$\frac{dT}{T} = \alpha dV$

$dQ = dU + dW$

$nCdT = nC_V dT + PdV$

$C = C_V + \frac{PdV}{ndT} = C_V + \frac{RT}{V}\frac{dV}{dT}$

$C = C_V + \frac{R}{\alpha V}$

12. C

Sol. At midpoint, let speed of particle is v

$\frac{1}{2}mv^2 = qE\frac{\pi R}{2}$

$a_r = \frac{v^2}{R}$, $a_t = \frac{qE}{m}$

$a_{net} = \sqrt{a_t^2 + a_r^2}$

13. C

Sol. Electric field due to line charge at distance r, $E = \frac{2k\lambda}{r}$

$\Delta V = -\int_R^{nR} \vec{E}\cdot d\vec{r} = -2k\lambda \int_R^{nR} \frac{dr}{r} = -2k\lambda \ell n(n)$

Gain in K.E. = loss in P.E.

$\Rightarrow \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2 = \frac{\lambda Q}{2\pi\varepsilon_0}\ell n(n)$, solve for V

14. C

Sol. For convex lens

$\frac{1}{v_1} - \frac{1}{-10} = \frac{1}{20} \Rightarrow v_1 = -20$

For plano-convex lens

$\frac{\mu}{\infty} - \frac{1}{-(d+20)} = \frac{\mu - 1}{R}$

$\frac{1}{f} = \frac{\mu - 1}{R} = \frac{1}{40}$

$\Rightarrow$ d = 20 cm

For More Material Join: @JEEAdvanced_2024

15. A

Sol. $d\sin\phi + \mu d\sin\theta = n\lambda$ (for constructive interference)

16. D

Sol. $\lambda = \frac{\ln 2}{24}$

$\frac{dv}{dt} = nv_0 - \lambda v$

$\Rightarrow \quad v = \frac{nv_0}{\lambda}\left(1 - e^{-\lambda t}\right)$

at t = 24 hrs $\Rightarrow v = \frac{12}{\ln 2} nv_0$

17. D

Sol. $I = \left(\frac{2}{(2n+1)\pi}\right)^2 I_0$

n = 2

$\Rightarrow I = \frac{I_0}{61}$(approx)

18. C

Sol. dV = volume of cone submerged, is nearly $\pi r^2 x$

$m = \frac{\rho}{2} \times 2 \times \frac{1}{3}\pi r^2 h = \frac{\rho \pi r^2 h}{3}$

$a = \frac{F}{m} = \frac{3g}{h}x$

$\Rightarrow \omega^2 = \frac{3g}{h}$

19. C

Sol. $f_{wall} = f\left(\frac{v - \frac{v}{k}}{v}\right)$

$f_{reflected} = f_{wall}\left(\frac{v}{v + \frac{v}{k}}\right)$

$f_r = f\left(\frac{v - \frac{v}{k}}{v + \frac{v}{k}}\right)$

$f_{beat} = f - f\frac{\left(v - \frac{v}{k}\right)}{\left(v + \frac{v}{k}\right)} = f\left[1 - \frac{1 - \frac{1}{k}}{1 + \frac{1}{k}}\right] = \frac{2f}{k+1}$

For More Material Join: @JEEAdvanced_2024

20. C

Sol. $I_C = \frac{10 - V_{CE}}{1000} = 5mA$

$\beta = \frac{I_C}{I_B};\ I_B = \frac{5\times10^{-3}}{100} = 5\times10^{-5}\,A$

$R_B = \frac{10 - V_{BE}}{I_B} = \frac{10-0}{5\times10^{-5}} = 2\times10^5\,\Omega$

SECTION – B

21. 14

Sol. $K.E._{max} = 5$ eV

Energy of incident photon = 7 eV

x_e (no. of e^- emitted per sec) = $\frac{i's}{e} = \frac{10^{-5}}{e}$

no. of photon incident per sec = $\frac{n_e}{0.5\times10^{-5}} = \frac{10^{-5}}{e\times0.5\times10^{-5}} = \frac{2}{e}$

incident power = (7 eV) $\frac{2}{e}$ = 14 W.

22. 25

Sol. $A = A_0e^{-\lambda t}$

$\frac{A_0 - A}{A_0} = 1 - e^{-\lambda t} = \frac{4}{100}$

$\Rightarrow$ t = 1 hr $\Rightarrow \lambda = 0.04$

$t_{mean} = \frac{1}{\lambda} = 25$ hrs.

23. 60

Sol. $2mv' = m2\sqrt{2}$

$v' = \sqrt{2}$

At position of maximum displacement gain in P.E. = loss in K.E.

$\frac{1}{2}kx^2 = \frac{1}{2}2mv'^2$

$\Rightarrow$ x = 0.2 m

Maximum displacement = $\sqrt{100^2 - 80^2} = 60$ cm

24. 7

Sol. Area under a-t graph gives change in velocity.

25. 3

Sol. No. of moles out flow (per minutes) = $\frac{10^5\times2.4\times10^{-3}}{R(300)}$ ...(i)

No. of moles of O_2 available = $\frac{1.50\times10^7\times16\times10^{-3}}{R(300)}$ ...(ii)

Time required to empty (from equation (ii) divide by equation (i)) = 10^3 minutes

For More Material Join: @JEEAdvanced_2024

26. 30

Sol. $P_1 = 75 + 5 = 80$ cm

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$\frac{(80-2x)(20+x)}{562.5} = \frac{80\times 20}{300}$$

Solving x = 10 cm

So, final length = 30 cm

27. 40

Sol. $$\frac{q}{C} + \frac{q}{2C} = 120V$$

$$\frac{3q}{2C} = 120$$

$$\Rightarrow \frac{q}{2C} = 40 \text{ V}$$

28. 12

Sol. Using Basic series parallel circuit simplification techniques

29. 150

Sol. $V_{AC} = E_3 = 30$ V

$V_{CB} = E_2$

$$\frac{E_2}{30} = \frac{V_{CB}}{V_{AC}} = \frac{3}{2} \Rightarrow E_2 = 45 \text{ V}$$

$V_{AB} = 30 + 45 = 75$ V

$R_{AB} = 10\Omega$

$$i_{AB} = \frac{75}{10} = 7.5 \text{ A}$$

$E_1 = 7.5(10+10) = 150$ V

30. 60

Sol. Image distance y = 3x

$$\frac{1}{f} = -\frac{1}{x} - \frac{1}{3x} \Rightarrow f = \frac{3x}{4}$$

$x' = x + 10 : y' = 2x'$

$$\frac{1}{x'} + \frac{1}{2x'} = \frac{1}{f} \Rightarrow f = \frac{2}{3}x'$$

$\Rightarrow x = 80 \Rightarrow f = 60$

For More Material Join: @JEEAdvanced_2024

Chemistry

PART – B

SECTION – A

31. D

Sol. $\left[Cu(NH_3)_4\right]^{2+}$ has square planar structure and is paramagnetic.

32. B

Sol. $ZnS + 4NaCN \rightleftharpoons Na^+\left[Zn(CN)_4\right]^{2-} + S^{2-}$

Water soluble

33. C

Sol. In I group only Li form nitride and all II group metals form nitride.

$Mg + N_2 \rightarrow Mg_3N_2 \xrightarrow{H_2O\ (moist)} Mg(OH)_2 + NH_3$

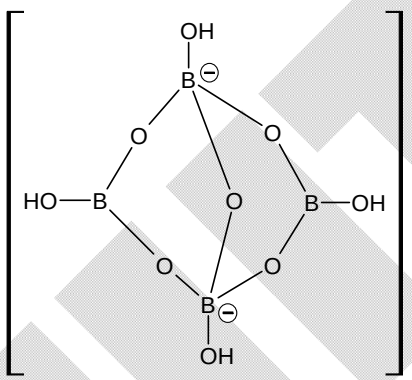
$Ca + N_2 \rightarrow Ca_3N_2 \xrightarrow{H_2O\ (moist)} Ca(OH)_2 + NH_3$

$Li + N_2 \rightarrow Li_3N \xrightarrow{H_2O\ (moist)} LiOH + NH_3$

$K + N_2 \rightarrow$ No reaction

34. B

Sol. In borax $(Na_2B_4O_7.10H_2O)$, among 10 water molecules 2 molecules are part of structure, i.e. $Na_2\left[B_4O_5(OH)_4\right].8H_2O$



35. C

Sol. NO_2 group, N in the ring and N in diazo (N = N) group do not respond Kjeldahl's test.

36. A

Sol. $FeSO_4 \longrightarrow Fe_2O_3 + SO_2 + SO_3$

(A) (D) (B) (C)

$Fe_2O_3 + 6HCl \longrightarrow 2FeCl_3 + 3H_2O$

(E)

37. A

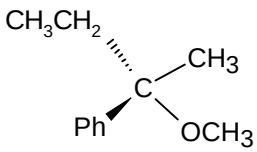
Sol. Larger the polarity of the sample, larger will be the attraction towards polar stationary phase.
Distance moved by A > B > C
Thus, polarity of A > B > C.
Thus, I is correct.

For More Material Join: @JEEAdvanced_2024

$$R_f = \frac{\text{Distance moved by solute}}{\text{Distance moved by solvent}}$$

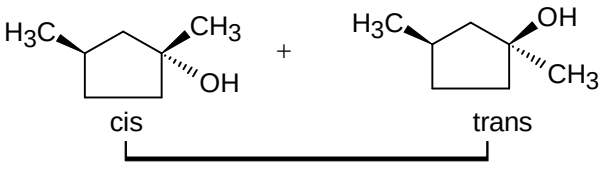
Thus, $R_f \propto$ distance moved by solute
Thus, R_f value of A > B > C.

38. B
Sol. (A)



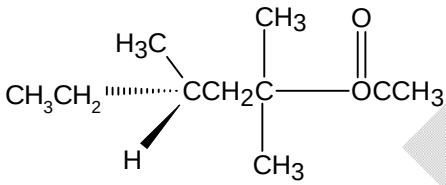
(Racemic), retention and inversion make racemic mixture

(B)



Diastereomers

(C)



(no chiral carbon at the leaving group.)

39. D
Sol.

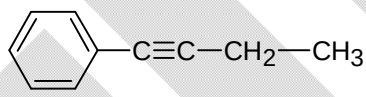
$$H_2C{=}CH_2 + HBr \longrightarrow H_3C{-}CH_2{-}Br$$

(A)

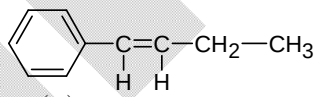
1° alkyl bromide

Thus, (A) is correct.
B is formed by S_N reaction.

B is

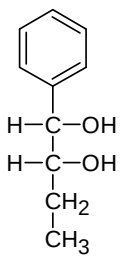


B $\xrightarrow{\text{Lindlar catalyst}}$



(C)
cis

$\xrightarrow{MnO_4^- / OH^-}$



(D)

For More Material Join: @JEEAdvanced_2024

Lindlar catalyst converts alkyne into cis-isomer. Thus, (C) is correct.

dil. MnO_4^- / OH^- and two (- OH) on alkene.

Thus, (B) is correct.

By syn-addtion.

Thus, (C) changes into cis isomer.

Thus, (D) is incorrect.

40. B

Sol. In $NH(SiH_3)_2$ electron density or lone pair at N-atom is involved in back bonding only with two empty 3d-orbitals of two silicon atoms while in $N(SiH_3)_3$ it is involved with three empty 3d-orbitals of three silicon atoms.

Strength of back bonding in $NH(SiH_3)_2$ is higher than in $N(SiH_3)_3$ hence N – Si bond length in $NH(SiH_3)_2$ is less than that of in $N(SiH_3)_3$.

41. A

Sol. Aqueous hydrolysis of CCl_4 is not possible due to absence of vacant orbital on carbon, while hydrolysis of $MgCl_2$, $SiCl_4$ and PCl_5 is possible. Extent of hydrolysis is proportional to the amount of positive formation charge present on the atomic site prone to attack of H_2O molecules.

$CCl_4 < MgCl_2 < AlCl_3 < SiCl_4 < PCl_5$

42. A

Sol. (A) $CuSO_4\,(aq.) \xrightarrow{\text{Excess of KCN}} \underset{(N)}{\left[Cu(CN)_4\right]^{3-}} + \underset{(O)}{(CN)_2}\uparrow$

43. D

Sol. (A) $2HNO_3 \xrightarrow{-H_2O} N_2O_5\,(\text{oxide of nitrogen})$

(B) $H_2SO_4 \xrightarrow{-H_2O} SO_3\,(\text{oxide of sulphur})$

(C) $2HClO_4 \xrightarrow{-H_2O} Cl_2O_7\,(\text{oxide of chloride})$

(D) $2H_2SO_4 \xrightarrow{-H_2O} H_2S_2O_7\,(\text{oxo}-\text{acid})$

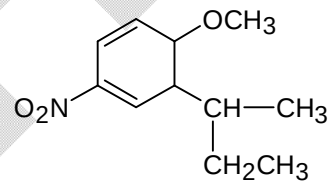
(A), (B) and (C) are oxides and not oxo-acids.

44. C

Sol. (A) $-SO_3H$ and $-\overset{\overset{O}{\|}}{C}CH_3$ are m-directing,

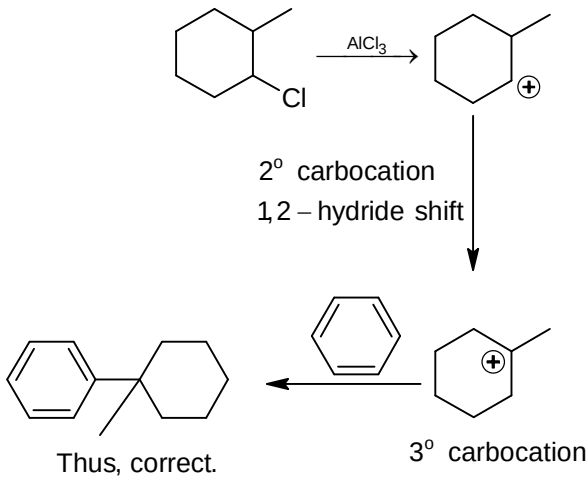
Thus, - Cl cannot go into o-position.

(B) $CH_3CH_2CH_2\overset{\oplus}{C}H_2$ changes to $CH_3CH_2\overset{\oplus}{C}HCH_3$ by 1, 2-hydride shift, thus product should be



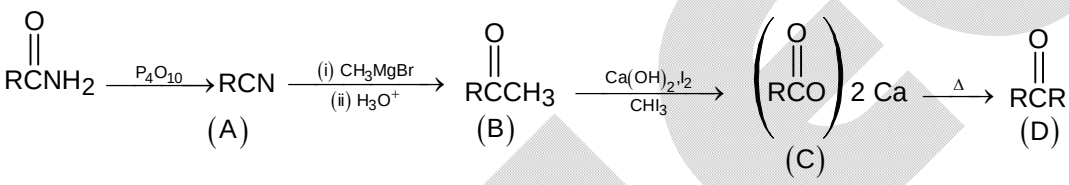
For More Material Join: @JEEAdvanced_2024

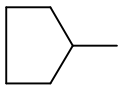
(C)

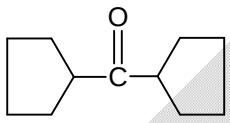


45. C

Sol.



R is  B gives iodoform test

Thus D is 

Thus, (C) is the correct answer.

46. C

Sol. Less stable intermediate are best hydride donor.

47. A

Sol. $\ln\frac{K_2}{K_1}=\frac{\Delta H}{R}\left[\frac{T_2-T_1}{T_1T_2}\right]$

Since, K increases with decrease of temperature and thus, $\Delta H = -ve$.

48. C

Sol. The pH of Zwitter ionic molecules do not depend upon their concentrations.

49. D

Sol. Veromal, amytal, Nembutal, luminal and seconal are derivatives of barbituric acid.

50. B

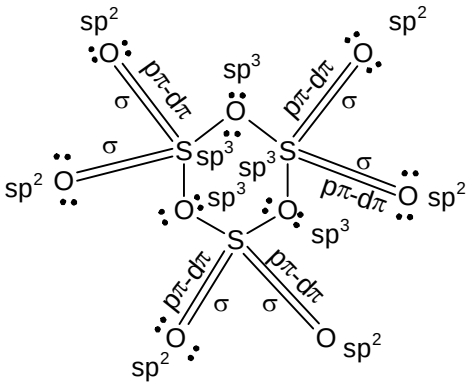
Sol. The small particle less than 10 micrometer in diameter pose the greatest problem.

For More Material Join: @JEEAdvanced_2024

SECTION – B

51. 1

Sol.



$p = 6,\ q = 6;\quad \frac{p}{q} = 1$

52. 4

Sol. Molecular orbital configuration for up to 14 electrons

$\sigma_{1s} < \sigma^*_{1s} < \sigma_{2s} < \sigma^*_{2s} < \pi_{2px} < \pi_{2py} < \sigma_{2pz} < \pi^*_{2px}$

$= \pi^*_{2py} < \sigma^*_{2pz}$

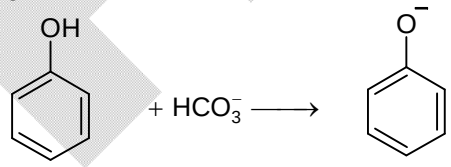
For more than 14 electrons

$\sigma_{1s} < \sigma^*_{1s} < \sigma_{2s} < \sigma^*_{2s} < \sigma_{2pz} < \pi_{2px} = \pi_{2py} < \pi^*_{2px}$

Number	Species	Total number	No. of electrons in	No. of electrons	Bond order
(i)	O_2^+	15	10	5	2.5
(ii)	CO	14	10	4	3
(iii)	B_2	10	6	4	1.0
(iv)	O_2^-	17	10	7	1.5
(v)	NO^+	14	10	4	3
(vi)	He_2^+	3	2	1	0.5
(vii)	C_2^{2+}	10	6	4	1.0
(viii)	CN^-	14	10	4	3.0
(ix)	N_2	15	10	5	2.5

53. 25

Sol.



+ H_2CO_3

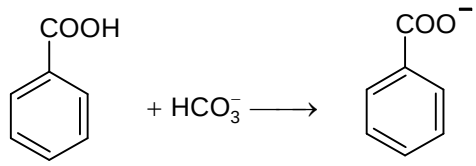
$pK_a = 6.38$

(Strong acid)

$pK_a = 10$

(Weak acid)

For More Material Join: @JEEAdvanced_2024



$pK_a = 4.20$ (Strong acid)

$pK_a = 6.38$ (Weak acid)

With phenol, a strong acid is formed. Thus, equilibrium favours backward reaction. With benzoic acid, a weak acid is formed, thus, CO_2 is due to it only.

CO_2 formed = 1 mol of benzoic acid

$PV = nRT$

$$\therefore \quad V = \frac{nRT}{P} = \frac{1 \times 0.0831 \times 300}{1}$$

$$= 24.93 \text{ dm}^3 \approx 25$$

54. 1

Sol. Acetic acid initially $= 2 \text{ g L}^{-1}$

Adsorbed on charcoal = α g

Thus, unadsorbed acetic acid $= (2-\alpha)$ g

Charcoal = 2g

$$\therefore \left(\frac{x}{m}\right) = k(C)^{1/n}$$

$\log k = -0.3010$

$k = 0.5$, slope $= 3.0$

$$\therefore \text{Slope} = \frac{1}{n} = \frac{1}{3}$$

$$\therefore \left(\frac{\alpha}{2}\right) = 0.5(2-\alpha)^{1/3}$$

$$\therefore \alpha^3 + \alpha - 2 = 1$$

$$\therefore (\alpha - 1)(\alpha^2 + \alpha + 2) = 0$$

$$\therefore \alpha = 1 \text{ g}$$

55. 2

Sol. $$pH = -\log 5 \times 10^{-10} + \log\left[\frac{5 \times V}{V+10} / \frac{10 \times 2}{V+10}\right]$$

or $$9 = -\log 5 \times 10^{-10} + \log\frac{V}{4}$$

$$\therefore V = 2 \text{ mL}.$$

56. 780

Sol. By Arrhenius equation,

For I. $k_1 = A_1 e^{-E_1/RT_1}$

For II. $k_2 = A_2 e^{-E_2/RT_2}$

$$\therefore \frac{k_1}{k_2} = \frac{A_1}{A_2} e^{(E_2 - E_1)/RT}$$

For More Material Join: @JEEAdvanced_2024

$$\frac{150\times10^{4}}{3.00\times10^{-4}}\left(\frac{A_1}{A_2}\right)e^{(15-30)/8.314\times10^{-3}\times600}$$

$$(0.5)=\left(\frac{A_1}{A_2}\right)e^{-3.007}$$

$$\therefore\left(\frac{A_1}{A_2}\right)=e^{-3.007}$$

$$=(0.5)\times20.227$$

= 10.1135

If equimolar qantities of the products are formed at temperature T, then $k_1 = k_2$.

$$\therefore \quad A_1e^{-E_1/RT}=A_2e^{-E_2/RT}$$

$$\therefore \quad \left(\frac{A_1}{A_2}\right)=e^{(E_1-E_2)/RT}$$

$$10.115=e^{(30-15)}/8.314\times10^{-3}\times T$$

$$2.3035\log10.1135=\frac{15}{8.314\times10^{-3}\times T}$$

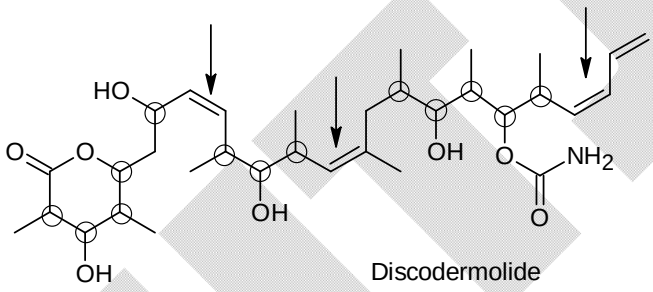
$\therefore T = 780$ K

57. 4

Sol. Pb, Cu, Ag *Hg evolve NO gas on their reaction with 20% HNO_3.

58. 16

Sol.



Discodermolide

Stereogenic centre = 13

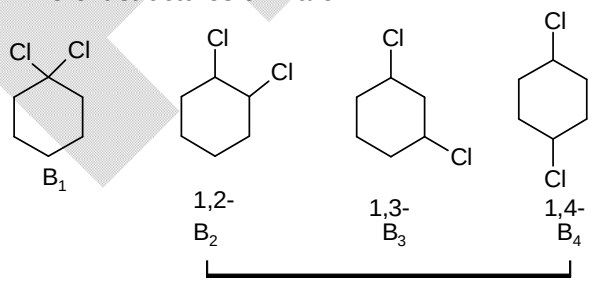
Having with tetrahedral character (encircled) (C = C) with stereogenic cente = 3 (Indicated by arrows)

Thus, maximum number of stereoisomers = $2^{16} = 2^{x}$

$\therefore x = 16$

59. 7

Sol. Different structures of B are



Also exist as pair of cis-and trans-

Thus, 1 + 6 = 7. Isomers (structural and geometrical).

For More Material Join: @JEEAdvanced_2024

60. 22

Sol. (C – Cl) bond length in $CH_3 – Cl$ = 178 pm

$l = 1.78 \times 10^{-8}$ cm

$e = \text{charge}$

$= 4.8 \times 10^{-10}$ esu.

$\therefore \mu_{CH_3-Cl}(\text{theoretical}) = e \times l$

$4.8 \times 10^{-10} \times 1.78 \times 10^{-8}$ esu – cm

$= 8.544 \times 10^{-18}$ esu – cm

= 8.544 Debye (D)

$(1 \times 10^{-8}$ esu – cm$) = 1$ D

$\mu_{CH_3-Cl}(\text{experimental}) = 1.87$ D

$$\% \text{ ionic character} = \frac{\text{Experimental } \mu}{\text{Theoretical } \mu} \times 100$$

$$= \frac{1.87}{8.544} \times 100$$

= 22%

For More Material Join: @JEEAdvanced_2024

Mathematics

PART – C

SECTION – A

61. C

Sol. We have

$$T = \frac{\tan\alpha(\cos\alpha+1)}{\cot\alpha(\sin\alpha+1)} = \frac{\tan^2\alpha(\cos\alpha+1)}{(\sin\alpha+1)} > 0$$

since α is not an integral multiple of $\frac{\pi}{2}$

62. A

Sol. We have $x = 2^{1/2}$, $y = 3^{1/3}$ and $z = 5^{1/5}$. Now $z^{10} = 25 < 32 = x^{10}$ and $x^6 = 8 < 9 = y^6$. Hence $z < x < y$.

63. D

Sol. Note that $0 < \cos\phi < 1 < \frac{\pi}{2}$. For all x in $\left(0,\frac{\pi}{2}\right)$, we have $\sin x < x$. Hence $\sin\cos\phi < \cos\phi$. Also, $\cos x$ is a decreasing function on this interval. Hence $\cos\phi < \cos\sin\phi$.

64. C

Sol. We have $f(1) = a - c$ and $f(2) = 4a - c$. Hence $a = \frac{1}{3}(f(2) - f(1))$ and $c = \frac{1}{3}(f(2) - 4f(1))$.

It follows that $f(3) = 9a - c = \frac{1}{3}(8f(2) - 5f(1))$

and $-1 = \frac{1}{3}(8(-1) - 5(-1)) \le f(3) \le \frac{1}{3}(8(5) - 5(-4)) = 20$.

65. B

Sol. Let $f(x) = \frac{1}{a^x - 1} + \frac{1}{2}$

Then $f(-x) = \frac{a^x}{1-a^x} + \frac{1}{2} = -f(x)$

Hence f(x) is an odd function. Since so is F(x) and G(x) = f(x) F(x), G(x) is an even function.

66. D

Sol. Let AB = x and consider the face BCD as the base. As x increases from 0, the altitude increase, and hence so does the volume. The volume reaches a maximum when the face ACD is perpendicular to the base. As x continues to increase towards 2, the altitude decreases, and hence so does the volume.

67. A

Sol. Clearly, we need three distinct edge lengths to make even one face non-isosceles. We can achieve the desired result with three by making two edges equal in length if and only if they are opposite to each other.

For More Material Join: @JEEAdvanced_2024

68. B

Sol. There are 7 points on C_7. At most 2 of the points on C_6 are on C_7. At most 4 of the points on C_5 are on C_6 or C_7. Hence M has at least 12 points. Draw four circles, no three concurrent and every two intersect at two points except for two which do not intersect at all. Call these two C_4 and C_5. Put all the points of intersections of these four circles into M, plus an additional point on each of C_5 and C_7. Then M has 12 points and C_i passes through exactly i points for $4 \le i \le 7$. It is easy to draw additional circles C_i for $1 \le i \le 3$, so that C_i passes through exactly i points already in M.

69. D

Sol. Note that $0 \le \arg z < 2\pi$ for all z. Hence (B) and (C) are both incorrect. Let $\alpha = \arg z = \frac{\pi}{6}$. Then

$\arg z^2 = \frac{\pi}{3}$ and $\overline{z}^2 = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3} \neq 2\left(\frac{\pi}{6}\right)$

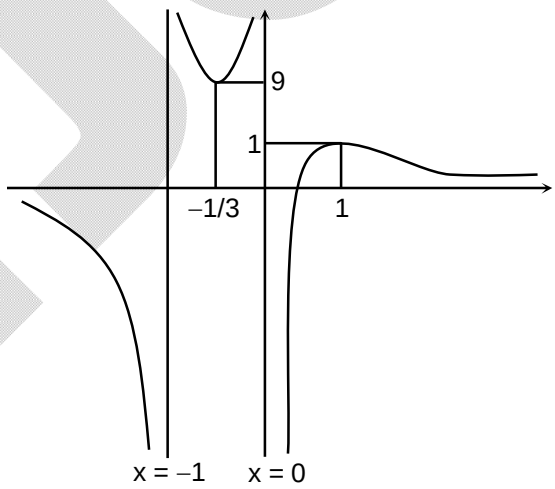
70. B

Sol. Note that $a^3 + 23 = (a - 1)(a(a + 1) + 1) + 24$ and $a(a + 1) + 1$ is odd. If $a^3 + 23$ is to be divisible by 24, so must $a - 1$ be. Now exactly one of $a - 1$, a and $a + 1$ is divisible by 3. If it is not $a - 1$, then $a(a + 1) + 1$ will not be divisible by 3 either, nor will $a^3 + 23$. It follows that $a - 1$ must be divisible by 24, and if this is the case, then $a^3 + 23$ will also be divisible by 24. Hence $a = 24k + 1 < 100$ for some non-negative integer k, and we have $k < 5$. It follows that there are 5 such values of a, namely, 1, 25, 49, 73 and 97.

71. D

Sol. $f(x) = \frac{3x-1}{x^2+x}$, $x \neq -1, 0$

$f'(x) = \frac{-(3x+1)(x-1)}{(x^2+x)^2}$



72. C

Sol. $f(x) = e^x + \frac{ex^2}{2}$

73. C

Sol. Make truth table.

74. D

Sol. For skew symmetric matrix

$a_{ij} = -a_{ji}$

$\Rightarrow$ only 6 position are to be filled

$\Rightarrow$ number of ways = 7^6

For More Material Join: @JEEAdvanced_2024

75. A

Sol. $\left(\frac{dx}{x}-\frac{dy}{y}\right)+\frac{x^2dy-y^2dx}{(x-y)^2}=0$

$\Rightarrow \left(\frac{dx}{x}-\frac{dy}{y}\right)+\left(\frac{dy/y^2-dx/x^2}{\left(\frac{1}{y}-\frac{1}{x}\right)^2}\right)=0$

$\Rightarrow \log\left|\frac{x}{y}\right|+\frac{xy}{x-y}=c$

76. D

Sol. Put $\frac{x}{a}=\frac{b}{y}$

77. B

Sol. $\cos\beta\pi=\frac{1}{2} \Rightarrow \beta=\pm\frac{1}{3}+2n,\ n\in I$

$\lim\limits_{x\to\pi}\frac{\sin\alpha x}{1-2\cos\beta x}=-\sqrt{3}$

$\Rightarrow \frac{(-1)^{\alpha}\cdot\alpha}{2\beta\cdot\frac{\sqrt{3}}{2}}=-\sqrt{3} \Rightarrow \alpha=1 \quad \beta=\frac{1}{3}$ for $(\alpha\beta)$ minimum.

78. D

Sol. $f'(x)=3x^2+2ax+b$

for f(x) to be increasing $4(a^2-3b)\le 0$

Req. probability $=\frac{16}{36}=\frac{4}{9}$

79. A

Sol. $\tan\frac{\alpha}{2}\cdot\tan\frac{\beta}{2}=\frac{1-e}{1+e}$ or $\frac{1+e}{1-e}$

80. B

Sol. Put $x = 1, \omega, \omega^2$ then add

where ω, ω^2 are non-real roots of unity.

SECTION – B

81. 1

Sol. mid-point of S_1 and S_2 is point of intersection of L_1 and L_2

$\Rightarrow (\alpha, \beta) = (1, 1)$

82. 8

Sol. $(x-1)^4 = 16$

$x = 3, -1 \quad 1+2i, 1-2i$

For More Material Join: @JEEAdvanced_2024

83. 5

Sol. Locus is a plane which is perpendicular to AB and passes through B(2, 3, 4)

$\Rightarrow x + 2y + 3z = 20$

84. 2

Sol. $\lim\limits_{x\to\infty}\sum\limits_{r=1}^{n}\left(\tan^{-1}\left(r+\frac{1}{2}\right)-\tan^{-1}\left(r-\frac{1}{2}\right)\right)$

$\Rightarrow \frac{\pi}{2}-\tan^{-1}\left(\frac{1}{2}\right)=\cot^{-1}\left(\frac{1}{2}\right)=\tan^{-1}2$

85. 8

Sol. $f(x) = 5 + [-4\sin x]$

86. 6

Sol. Put $x - 2 \to x$ and $y + 1 \to y$

$1 \le |x| + |y| \le 2$

87. 12

Sol. Put $x+\sqrt{x^2+1} \to t$

$I_n = \frac{n}{n-1}$

88. 6

Sol. Since 2, α_1, α_2 ……., α_{20}, 6 are in H.P.

$\alpha_{18} = \frac{14}{3}$

Also 2, β_1, β_2, …… , β_{20}, 6 are in A.P. $\beta_3 = \frac{18}{7}$

So, $\alpha_{18}.\beta_3 = 12$

89. 9

Sol. $\left|\vec{a}-\vec{b}\right|=2\sin\frac{\theta}{2}$ and $\left|\vec{a}+\vec{b}\right|=2\cos\frac{\theta}{2}$

so $-\sqrt{85}\le 3\left|\vec{a}-\vec{b}\right|+\frac{7}{2}\left|\vec{a}+\vec{b}\right|\le\sqrt{85}$

90. 27

Sol. $\int e^x\cdot f(x)dx = e^x\left(f(x)-f'(x)+f''(x).....\right)+c$ where f(x) is a polynomial.

For More Material Join: @JEEAdvanced_2024